

A PROJECT TITLE

ON

Fertilizer Requirement Prediction Using Regularized Logistic Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: **Fertilizer Requirement Prediction Using Regularized Logistic Regression**

Abstract:

Agriculture plays a vital role in ensuring food security and economic growth. Proper fertilizer application is essential for improving crop productivity and maintaining soil health. However, selecting the right fertilizer based on soil and environmental conditions is often difficult for farmers. This project presents a machine learning approach to predict the most suitable fertilizer using **Regularized Logistic Regression**.

The system uses agricultural data such as soil nitrogen, phosphorus, potassium levels, pH value, moisture content, crop type, rainfall, and temperature. Before training the model, the dataset is pre-processed by handling missing values, encoding categorical variables, normalizing numerical features, and removing inconsistencies. The processed data is then divided into training and testing datasets.

Regularized Logistic Regression applies L1 or L2 regularization to reduce overfitting and improve the model's prediction performance. The trained model classifies the appropriate fertilizer for a given set of soil and crop conditions. Model performance is evaluated using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

The proposed system provides farmers with reliable fertilizer recommendations, helping them increase crop yield, reduce fertilizer wastage, lower production costs, and promote environmentally sustainable farming practices. This project demonstrates the effectiveness of machine learning in modern precision agriculture.

1. INTRODUCTION:

Agriculture is one of the most important sectors contributing to the economy and food production. Fertilizers supply essential nutrients required for healthy plant growth. Applying incorrect fertilizers or excessive quantities may reduce crop yield, increase costs, and negatively affect soil quality and the environment.

This project, "**Fertilizer Requirement Prediction Using Regularized Logistic Regression**," aims to recommend the most suitable fertilizer based on soil and environmental parameters. The system uses historical agricultural data containing soil nutrient values, crop information, pH levels, rainfall, temperature, humidity, and moisture content.

The collected data undergoes preprocessing to improve quality before model training. Regularized Logistic Regression is then used to classify fertilizer recommendations while minimizing overfitting through regularization techniques. The trained model predicts fertilizer requirements with improved accuracy and reliability.

The proposed system supports farmers and agricultural experts in making data-driven decisions for fertilizer management. It improves crop productivity, reduces unnecessary fertilizer usage, lowers farming costs, and encourages sustainable agricultural practices.

2. SOFTWARE REQUIREMENTS:

Operating System

- Windows 10/11
- Linux
- macOS

Programming Language

- Python 3.9 or above

Development Environment

- Jupyter Notebook
- Google Colab
- Visual Studio Code (VS Code)

Libraries and Frameworks

- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

Dataset

- Fertilizer Prediction Dataset (CSV)

Machine Learning Algorithm

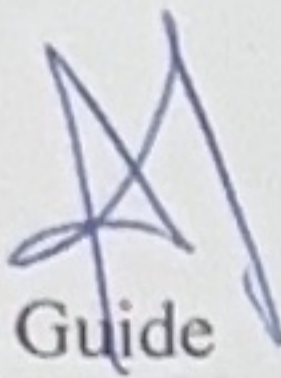
- Regularized Logistic Regression (L1/L2)

Visualization Tools

- Matplotlib

3. SOFTWARE REQUIREMENTS:

- ☐ **Operating System:** Windows 10/11 or Linux
- ☐ **Programming Language:** Python 3.x
- ☐ **IDE:** Jupyter Notebook / Google Collab / VS Code
- ☐ **Libraries:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- ☐ **Dataset:** Fertilizer Prediction Dataset
- ☐ **Machine Learning Algorithm:** Regularized Logistic Regression (L1/L2)
- ☐ **Version Control:** Git (Optional)



Signature of the Guide



Course Instructor